GRE Quantitative Reasoning

모의고사 Set 14 — 문제지

최신 문제(뒷번호) · S2 고난도

구성
Section 1: 12문항 / 21분
Section 2: 15문항 / 26분
합계 27문항 / 47분 · 계산기 사용 가능

대소비교(Quantitative Comparison) 선택지 — 모든 QC 문제 공통
(A) Quantity A is greater. (B) Quantity B is greater.
(C) The two quantities are equal. (D) The relationship cannot be determined from the information given.

안내

- Indicate all (복수선택) 문제는 해당하는 것을 모두 고르세요. 모두 맞아야 정답입니다.
- 주관식 분수 답은 약분하지 않아도 됩니다.
- 문제 이미지 속 번호는 문제집 원번호입니다. 문항 번호는 보라색 'Question N'을 기준으로 합니다.
- 정답과 해설은 별도 파일(Set 14 해설지)에 있습니다.

답안 기록표

S1	1	2	3	4	5	6	7	8	9	10	11	12
답												

S2	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
답															

Section 1 12 Questions · 21 Minutes

Question 1 대소비교

1142. A circle has an arc of length 16, which is 20 percent of the circumference of the circle.
Quantity A: The area of the circle
Quantity B: 1600/ π

(A) Quantity A is greater.
(B) Quantity B is greater.
(C) The two quantities are equal.
(D) The relationship cannot be determined from the information given.

Question 2 대소비교

662. x - y =5
Quantity A: x^2-y^2
Quantity B: 5

(A) Quantity A is greater.
(B) Quantity B is greater.
(C) The two quantities are equal.
(D) The relationship cannot be determined from the information given.

Question 3 대소비교

805. 100 < a < b < 1000
Quantity A: (1/a) − (1/b)
Quantity B: (1/100) – (1/1000)

(A) Quantity A is greater.
(B) Quantity B is greater.
(C) The two quantities are equal.
(D) The relationship cannot be determined from the information given.

Question 4 대소비교

930. n and k are integers, n>k>1
Quantity A: (n−k)!
Quantity B: n!−k!

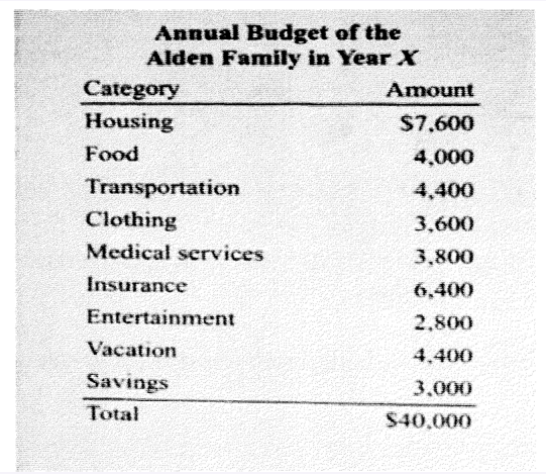
(A) Quantity A is greater.
(B) Quantity B is greater.
(C) The two quantities are equal.
(D) The relationship cannot be determined from the information given.

Question 5 객관식

1040. What percent of 1/2 is 2/3?
A 33(1/3)%
B 75%
C 116(2/3)%
D 133(1/3)%
E 175%

A B C D E

Questions 6 to 8 are based on the following data.



Annual Budget of the Alden Family in Year *X*

Category	Amount
Housing	$7,600
Food	4,000
Transportation	4,400
Clothing	3,600
Medical services	3,800
Insurance	6,400
Entertainment	2,800
Vacation	4,400
Savings	3,000
Total	$40,000

Question 6 객관식

551. If 20 percent of the amount budgeted for vacation had instead been added to the amount budgeted for entertainment, which of the following is closest to the percent by which the amount budgeted for entertainment would have increased?
A10%
B15%
C20%
D25%
E30%

A B C D E

Question 7 객관식

552. The three children in the Alden family contributed 30 percent of the amount budgeted for savings. The oldest child contributed $150 more than the youngest child, and the middle child contributed $60 more than the youngest child. What was the median of the amounts that the children contributed to savings ?
A$230
B$270
C$290
D$300
E$380

A B C D E

Question 8 객관식

553. In each of the years after year X, the Alden family budgeted $200 more for savings than it budgeted for savings the previous year. What is the total amount that the Alden family budgeted for savings over the 10 years beginning with year X?
A$30,000
B$32,000
C$36,000
D$39,000
E$42,000

A B C D E

Question 9 객관식

701. The discounted price of a certain suit is 20 percent less than the original price of the suit. If the discounted price of the suit plus a sales tax of 5 percent of the discounted price equals $67.20, what was the original price of the suit?
A$70.50
B$73.90
C$76.00
D$79.80
E$80.00

A B C D E

Question 10 주관식

694. There are 5!, or 120, ways of arranging 5 different solid-colored flags side by side. If the colors of the flags are red, blue, yellow, green, and orange, how many of those arrangements have either the red flag or the blue flag in the middle position?

Question 11 복수선택

922. If the ones digit of 7^n is 9, which of the following could be the value of n? Indicate all such values.
A100
B101
C102
D103
E104
F105
G106

Indicate all that apply. A B C D E F G

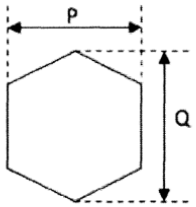
Question 12 객관식

938. $(2.82\times10^{-51}-3.96\times10^{-49})=$
A-3.9318×10^{-49}
B-1.7804×10^{-51}
C-1.14×10^{-100}
D1.7804×10^{-51}
E3.9318×10^{-49}

A B C D E

Section 2 15 Questions · 26 Minutes

Question 1 대소비교



471.The hexagon is both equilateral and equiangular.
Quantity A: P
Quantity B: Q

(A) Quantity A is greater.
(B) Quantity B is greater.
(C) The two quantities are equal.
(D) The relationship cannot be determined from the information given.

Question 2 대소비교

1096. 1, 2, 4, 8, 16.....
The first five terms of an infinite sequence are shown above. Each term after the first term is 2 times the preceding term.
n is an odd integer greater than 50.
Quantity A: The average (arithmetic mean) of the first n terms of the sequence
Quantity B: The median (arithmetic mean) of the first n terms of the sequence

(A) Quantity A is greater.
(B) Quantity B is greater.
(C) The two quantities are equal.
(D) The relationship cannot be determined from the information given.

Question 3 대소비교

438. The four-digit integer 8,82X is a multiple of 9, where X is the units digit.
Quantity A: X
Quantity B: 2

(A) Quantity A is greater.
(B) Quantity B is greater.
(C) The two quantities are equal.
(D) The relationship cannot be determined from the information given.

Question 4 대소비교

1155. In the distribution of measurements of the variable x, the mean is 56 and the measurement r lies between the 65th and 70th percentiles. In the distribution of measurements of the variable y, the mean is 56 and the measurement t lies between the 75th and 80th percentiles.
Quantity A: r
Quantity B: t

(A) Quantity A is greater.
(B) Quantity B is greater.
(C) The two quantities are equal.
(D) The relationship cannot be determined from the information given.

Question 5 대소비교

743.

Chemistry Test Scores for Two Classes		
	Median Score	Number of Scores
Class A	74	50
Class B	78	65

Quantity A: The median of the 115 scores
Quantity B: 76

(A) Quantity A is greater.
(B) Quantity B is greater.
(C) The two quantities are equal.
(D) The relationship cannot be determined from the information given.

Question 6 객관식

345. A computer identification code on a certain machine consists of 2 letters from an alphabet of 26 letters, followed by 2 digits from the digits 0 to 9. The two digits must be different. How many identification codes are possible?
A26*9
B26*10
C26*10*9
D26^2*9
E26^2*10*9

(A) (B) (C) (D) (E)

Question 7 객관식

447. If n is a positive odd integer and $k=n^3+2n$, what is the value of $(-1)^k-(-1)^{k+1}$?
A-2
B-1
C0
D1
E2

(A) (B) (C) (D) (E)

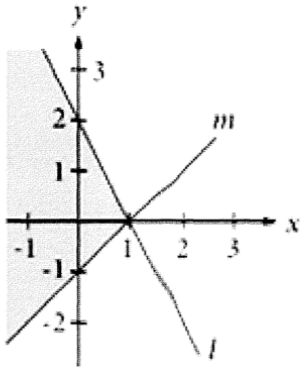
Question 8 객관식

1184. For a list of 10 different numbers, the average (arithmetic mean) of the numbers is 22 and the range is 50. If both the least number and the greatest number are removed from the list, then the average of the remaining numbers is 20. What is the greatest number in the list of 10 numbers?
A.55
B.52
C.45
D.42
E.37

Ⓐ Ⓑ Ⓒ Ⓓ Ⓔ

Question 9 복수선택

463. In the xy-plane above, both the x-intercept and the y-intercept of lines l and m are integers.



The coordinates (x, y) of each point in the shaded region satisfy which two of the following inequalities? Indicate BOTH of the inequalities.
A $y \leq x + 1$
B $y \leq x - 1$
C $y \leq 2x + 2$
D $y \leq -2x + 2$
E $y \geq x + 1$
F $y \geq x - 1$
G $y \geq 2x + 2$
H $y \geq -2x + 2$

Indicate all that apply. [A] [B] [C] [D] [E] [F] [G] [H]

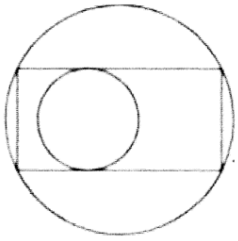
Question 10 주관식

808. On a radio call-in show, the 6th caller and every 6th caller thereafter were given a T-shirt. Also, the 15th caller and every 15th caller thereafter were given movie tickets. A total of 78 people called the show, all at different times. How many of the callers were given neither a T-shirt nor movie tickets? _____ callers

[ ] callers

Question 11 객관식

1154.



The figure above shows a rectangle inscribed in a large circle. Inside the rectangle is a small circle of radius 2 that is tangent to two sides of the rectangle. If the length of the rectangle is twice its width, what is the area of the large circle?
A.12 π √ 3
B.16 π √ 2
C.16 π
D.18 π
E.20 π

Ⓐ Ⓑ Ⓒ Ⓓ Ⓔ

Question 12 객관식

1010. Which of the following is a value of x for which $x^{12}-x^{10}+x^{8}-x^{6}>0$?
A-2
B-1
C−1/2
D1/2
E1

Ⓐ Ⓑ Ⓒ Ⓓ Ⓔ

Question 13 복수선택

351. If n is a positive integer greater than 1, then $n(n^2-1)$ must be a multiple of which of the following integers? Indicate all such numbers.
A2
B3
C4
D5
E6

Indicate all that apply. [A] [B] [C] [D] [E]

Question 14 객관식

912. If x is a positive integer such that the units digit of x^3 is 3, what is the units digit of x^{15}?
A1
B3
C5
D7
E9

Ⓐ Ⓑ Ⓒ Ⓓ Ⓔ

Question 15 객관식

1330. $(5^3)w+(5^2)x+5y+z=264$

In the equation shown, w, x, y and z are integers that are non-negative integers and each is less than 5. What is the value of w+x+y+z?

A.5
B.6
C.8
D.10
E.12

A B C D E